Multi-Fidelity Inference of Electrochemical Kinetics: A Surrogate-Accelerated PDE-Constrained Optimization Pipeline

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March 2026

Abstract

We present a multi-fidelity inference pipeline for recovering four Butler–Volmer kinetic parameters governing a two-reaction oxygen reduction mechanism from noisy current–voltage measurements. The pipeline combines neural-network ensemble surrogates for rapid global exploration with PDE-constrained optimization on a Poisson–Nernst–Planck–Butler–Volmer forward model for high-accuracy refinement. A seven-phase architecture (five surrogate phases followed by two PDE phases) achieves $< 5.3\%$ error on all four parameters simultaneously from data corrupted with 2% Gaussian noise, overcoming a persistent $k_{0,1}$–$k_{0,2}$ anti-correlation that makes joint recovery of both rate constants challenging. The key architectural insight is a continuation strategy: a shallow-voltage PDE phase selects the correct optimization basin before a full-cathodic PDE phase performs final refinement. Surrogate phases contribute only 2% of total compute cost yet provide essential warm-starting that reduces PDE phase wall-clock time by 30%.

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1 Introduction

Electrochemical kinetics govern the performance of fuel cells, batteries, and electrolyzers. Accurate determination of kinetic rate constants and charge-transfer coefficients is essential for rational catalyst design and predictive modeling. The oxygen reduction reaction (ORR) on rotating disk electrodes (RDEs) is a canonical test system, with well-defined mass transport that enables isolation of kinetic parameters from measured current–voltage (I–V) curves.

The inverse problem. We consider a two-reaction ORR mechanism:

$$\mathrm{R}_1: \quad \mathrm{O_2} + 2\mathrm{H}^+ + 2e^- \rightarrow \mathrm{H_2O_2}, \tag{1}$$

$$\mathrm{R}_2: \quad \mathrm{H_2O_2} + 2\mathrm{H}^+ + 2e^- \rightarrow 2\mathrm{H_2O}. \tag{2}$$

Given noisy I–V measurements, we seek four kinetic parameters: the heterogeneous rate constants $k_0^{(1)}$, $k_0^{(2)}$ and the cathodic charge-transfer coefficients α_1, α_2.

Challenge. The rate constant $k_0^{(2)}$ is approximately 24× smaller than $k_0^{(1)}$, so Reaction 2's contribution to total current is weak at moderate overpotentials. Its signal is buried in measurement noise and obscured by a strong anti-correlation between $k_0^{(1)}$ and $k_0^{(2)}$: improving one at the expense of the other can reduce the objective without improving physical accuracy.

Contribution. We introduce a seven-phase multi-fidelity pipeline that achieves $< 5.3\%$ maximum error across all four parameters. The architecture comprises:

1. Five surrogate phases using a neural-network ensemble for global exploration (cascade, multi-start, and joint optimization),
2. Two PDE-constrained optimization phases that exploit the full physics of the Poisson–Nernst–Planck system with Butler–Volmer boundary conditions.

The key insight is that a shallow-voltage PDE phase (P1) serves as a *basin selector* for the full-cathodic PDE phase (P2), analogous to a continuation method in nonlinear optimization.

Outline. Section 2 defines the electrochemical system and governing equations. Section 3 describes the finite-element forward solver. Section 4 formulates the inverse problem. Section 5 presents the surrogate models. Section 6 details the seven-phase pipeline architecture. Section 7 reports numerical results. Section 8 discusses computational optimizations. Section 9 provides analysis and context. Section 10 concludes.

2 Problem Setup

2.1 Electrochemical System

We model oxygen reduction on a rotating disk electrode in acidic electrolyte (pH 4, 0.1 M $\mathrm{HClO_4}$). The ORR proceeds via the two-electron pathway (1)–(2), producing hydrogen peroxide as an intermediate. Four ionic species participate:

Table 1: Species definitions for the four-species charged system.

Index i	Species	Charge z_i	D_i (m^2/s)	c_i^{bulk} (mol/m^3)	$\hat{c}_i^0$
0	$\mathrm{O_2}$	0	1.90×10^{-9}	0.5	1.0
1	$\mathrm{H_2O_2}$	0	1.60×10^{-9}	0.0	0.0
2	H^+	+1	9.311×10^{-9}	100.0	0.2
3	$\mathrm{ClO_4^-}$	−1	1.792×10^{-9}	100.0	0.2

Charge neutrality is maintained: $[H^+] = [ClO_4^-] = 0.1\,\mathrm{M}$. The physical constants are drawn from the Mangan2025 catalysis study and are listed in Table 2.

Table 2: Physical constants and kinetic parameters.

Symbol	Description	Value	Unit
F	Faraday constant	96,485.33	C/mol
R	Gas constant	8.3145	J/(mol K)
T	Temperature	298.15	K
V_T	Thermal voltage RT/F	0.02569	V
ε_r	Relative permittivity (water)	78.5	—
ε_0	Vacuum permittivity	8.854×10^{-12}	F/m
n_e	Electrons per reaction	2	—
$k_0^{(1)}$	Rate constant, R_1	2.4×10^{-8}	m/s
$k_0^{(2)}$	Rate constant, R_2	1.0×10^{-9}	m/s
α_1	Transfer coefficient, R_1	0.627	—
α_2	Transfer coefficient, R_2	0.500	—

2.2 Governing Equations

The species concentrations $c_i(\boldsymbol{x}, t)$ and electric potential $\phi(\boldsymbol{x}, t)$ satisfy the coupled Poisson–Nernst–Planck (PNP) system. In nondimensional form:

Nernst–Planck transport. For each species $i = 0, \ldots, N_s - 1$:

$$\frac{\partial \hat{c}_i}{\partial \hat{t}} = \nabla \cdot \left[\hat{D}_i\big(\nabla \hat{c}_i + z_i\, \hat{c}_i\, \nabla \hat{\phi} + \hat{c}_i\, \nabla \mu_{\text{steric}}\big)\right], \tag{3}$$

where $\hat{D}_i = D_i/D_{\text{ref}}$ is the dimensionless diffusivity and the steric (Bikerman) chemical potential is

$$\mu_{\text{steric}} = \ln\big(1 - \textstyle\sum_j a_j \hat{c}_j\big), \tag{4}$$

with $a_j = 0.01$ the steric parameter for all species.

Poisson equation.

$$-\hat{\varepsilon}\, \nabla^2 \hat{\phi} = \sum_{i=0}^{N_s-1} z_i\, \hat{c}_i, \tag{5}$$

where $\hat{\varepsilon} = (\lambda_D/L_{\text{ref}})^2$ is the squared Debye-to-domain length ratio.

Butler–Volmer kinetics. At the electrode surface, the flux of species i due to reaction j is:

$$\mathcal{R}_j = \hat{k}_0^{(j)} \Big[\hat{c}_{\text{cat},j}\, \exp\big(-\alpha_j\, \hat{\eta}\big) - \hat{c}_{\text{ref},j}\, \exp\big((1 - \alpha_j)\, \hat{\eta}\big)\Big], \tag{6}$$

where $\hat{\eta} = (\phi_{\text{applied}} - E_{\text{eq}})/V_T$ is the dimensionless overpotential, $\hat{c}_{\text{cat},j}$ is the surface concentration of the cathodic reactant, and $\hat{c}_{\text{ref},j}$ is a reference concentration for the anodic branch. Reaction 2 is irreversible (no anodic term). The net flux boundary condition for species i on the electrode is:

$$-\hat{D}_i\, \nabla \hat{c}_i \cdot \boldsymbol{n} = \sum_j s_{ij}\, \mathcal{R}_j, \tag{7}$$

where s_{ij} is the stoichiometric coefficient of species i in reaction j (see Table 3).

Table 3: Stoichiometric coefficients s_{ij} for the two-reaction mechanism.

	O_2	H_2O_2	H^+	ClO_4^-
R_1: $O_2 \to H_2O_2$	-1	$+1$	-2	0
R_2: $H_2O_2 \to H_2O$	0	-1	-2	0

2.3 Nondimensionalization

We introduce reference scales to obtain the dimensionless system (3)–(5). The reference quantities and derived scales are listed in Table 4.

Table 4: Reference scales and derived nondimensional groups.

Scale	Symbol	Expression	Typical value
Length	L_{ref}	Domain height	100 µm
Diffusivity	D_{ref}	D_{O_2}	$1.9 \cdot 10^{-9}\,\text{m}^2/\text{s}$
Concentration	c_{ref}	$c_{O_2}^{\text{bulk}}$	$0.5\,\text{mol}/\text{m}^3$
Potential	V_T	RT/F	25.69 mV
Time	t_{ref}	$L_{\text{ref}}^2/D_{\text{ref}}$	5.26 s
Velocity	κ_{ref}	$D_{\text{ref}}/L_{\text{ref}}$	$1.9 \cdot 10^{-5}\,\text{m}/\text{s}$
Current density	I_{ref}	$n_e F D_{\text{ref}} c_{\text{ref}}/L_{\text{ref}}$	$1.83\,\text{mA}/\text{cm}^2$
Debye length	λ_D	$\sqrt{\varepsilon_r \varepsilon_0 V_T/(F\, c_{\text{bulk}})}$	$\sim$3 nm
Debye ratio	$\hat{\varepsilon}$	$(\lambda_D/L_{\text{ref}})^2$	$\sim 9 \times 10^{-8}$

The rate constants are nondimensionalized as $\hat{k}_0^{(j)} = k_0^{(j)}/\kappa_{\text{ref}}$, giving $\hat{k}_0^{(1)} \approx 1.26$ and $\hat{k}_0^{(2)} \approx 0.053$. The current density scale I_{ref} converts dimensionless species flux to physical units via $I_{\text{CD}} = I_{\text{ref}} \times$ (dimensionless flux).

2.4 Domain and Boundary Conditions

The computational domain is a two-dimensional rectangle $\Omega = [0, L_x] \times [0, L_y]$ with graded mesh (see Section 3.1 for discretization details). The boundary is partitioned as shown in Figure 1.

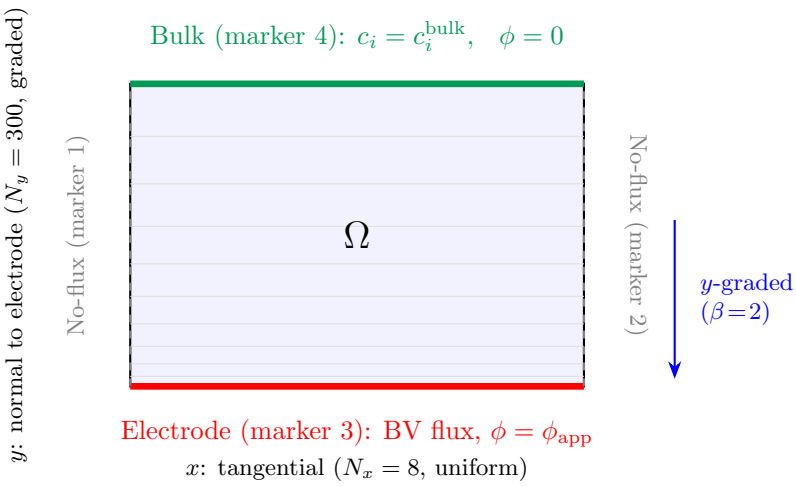

Figure 1: Computational domain representing the diffusion layer above a rotating disk electrode. The bottom boundary (marker 3) is the electrode surface with Butler–Volmer flux boundary conditions; the top boundary (marker 4) imposes bulk Dirichlet concentrations and ground potential; left and right boundaries (markers 1,2) are no-flux (symmetry). The mesh is graded in y with $\beta = 2$, clustering elements near the electrode to resolve steep concentration gradients and the Debye screening layer.

2.5 Observables

Two observables are extracted from each forward solve at overpotential η:

Current density I_{CD}. The total Faradaic current density, integrated over the electrode surface:

$$I_{\text{CD}}(\eta) = I_{\text{ref}} \sum_j \int_{\Gamma_{\text{elec}}} \mathcal{R}_j \, \mathrm{d}s. \tag{8}$$

This captures the overall reduction current (dominated by R_1).

Peroxide current I_{PC}. The current associated with net peroxide production:

$$I_{\text{PC}}(\eta) = I_{\text{ref}} \int_{\Gamma_{\text{elec}}} \left(\mathcal{R}_1 - \mathcal{R}_2\right) \mathrm{d}s. \tag{9}$$

Since $\mathcal{R}_1$ produces H_2O_2 and $\mathcal{R}_2$ consumes it, I_{PC} is sensitive to the ratio $k_0^{(2)}/k_0^{(1)}$ and is essential for identifiability of the four-parameter system. Without I_{PC}, the $k_0^{(1)}$–$k_0^{(2)}$ pair is poorly determined.

Experimental measurement. Both observables are accessible in a single experiment using a rotating ring-disk electrode (RRDE) [2]. An RRDE consists of a small *disk* (the catalyst surface) surrounded by a concentric *ring*, both submerged in solution and spinning. The disk voltage is swept while its current is recorded—this is I_{CD}, capturing the total electrochemical activity (both $O_2 \to H_2O_2$ and $H_2O_2 \to H_2O$). The ring is held at a fixed potential that oxidizes any H_2O_2 drifting outward from the disk; its current, after dividing by a known geometric correction factor (collection efficiency N), gives I_{PC}. Sweeping the disk potential therefore produces both I–V curves simultaneously—exactly the pair of observables the inverse problem needs to disentangle the two reaction rate constants. This experimental grounding motivates the two-observable objective function (18): the disk current constrains the dominant reaction (R_1) while the ring current constrains the secondary reaction (R_2).

3 Forward Solver

The forward solver computes steady-state solutions of the PNP–BV system (3)–(7) at a sequence of overpotentials, producing I–V curves for both observables. It is implemented in Firedrake, a finite-element framework built on PETSc.

3.1 Finite Element Discretization

Mesh. The domain is discretized with a structured rectangle mesh: $N_x = 8$ elements in the tangential direction (uniform) and $N_y = 300$ elements in the normal direction with power-law grading:

$$y_k = \left(\frac{k}{N_y}\right)^{\beta}, \qquad k = 0, \ldots, N_y, \tag{10}$$

with $\beta = 2$. This clusters nodes near the electrode ($y = 0$) to resolve thin boundary layers and the Debye screening region.

Function space. We use a mixed function space consisting of $N_s + 1$ copies of continuous Galerkin elements of degree 1 (CG_1):

$$W = \underbrace{V_h \times \cdots \times V_h}_{N_s \text{ species}} \times V_h^{(\phi)}, \qquad V_h = \mathrm{CG}_1(\mathcal{T}_h). \tag{11}$$

The solution vector $U = (c_0, c_1, \ldots, c_{N_s-1}, \phi) \in W$ combines all species concentrations and the electric potential.

Weak form. The variational formulation for test function $v = (v_0, \ldots, v_{N_s-1}, w) \in W$ reads:

$$\begin{aligned} F(U; v) = \sum_{i=0}^{N_s-1} \int_\Omega \frac{c_i - c_i^{\mathrm{old}}}{\Delta t} v_i \,\mathrm{d}x + \sum_{i=0}^{N_s-1} \int_\Omega \hat{D}_i \big(\nabla c_i + z_i\, c_i \nabla\phi + c_i \nabla\mu_{\mathrm{steric}}\big) \cdot \nabla v_i \,\mathrm{d}x \\ - \sum_{i,j} s_{ij} \int_{\Gamma_{\mathrm{elec}}} \mathcal{R}_j\, v_i \,\mathrm{d}s + \hat{\varepsilon} \int_\Omega \nabla\phi \cdot \nabla w \,\mathrm{d}x - \int_\Omega \Big(\sum_i z_i c_i\Big) w \,\mathrm{d}x = 0. \end{aligned} \tag{12}$$

The system is solved with Newton's method using PETSc's SNES (Scalable Nonlinear Equations Solvers) with MUMPS direct factorization.

3.2 Pseudo-Transient Continuation

Direct Newton iteration on the steady-state residual often diverges for large overpotentials due to the exponential nonlinearity of the Butler–Volmer terms. We embed the steady-state problem in fictitious time-stepping:

$$\frac{1}{\Delta t_{\mathrm{ptc}}} M(U^{n+1} - U^n) + F(U^{n+1}) = 0, \tag{13}$$

where M is the mass matrix. As $\Delta t_{\mathrm{ptc}} \to \infty$, (13) reduces to the steady-state problem $F(U) = 0$.

Adaptive time-stepping. The pseudo-time step is adapted using the Switched Evolution Relaxation (SER) strategy:

$$\Delta t_{\mathrm{ptc}}^{n+1} = \Delta t_{\mathrm{ptc}}^{n} \cdot \frac{\left\|r^{n-1}\right\|}{\max(\|U^n - U^{n-1}\|\,,\, 10^{-30})} \cdot \gamma, \tag{14}$$

where $\gamma = 1.5$ is the growth factor and the step is clamped at $\Delta t_{\max} = 10^6$. The initial step is $\Delta t_0 = 0.01$, which provides Picard-like regularization. Steady state is declared when 4 consecutive steps satisfy $\left\|U^{n+1} - U^n\right\| / \|U^n\| < 10^{-5}$.

3.3 Voltage Sweep via Continuation

An I–V curve requires solving at multiple overpotentials $\eta_1, \eta_2, \ldots, \eta_M$. Rather than solving each independently, we use *voltage continuation*: the converged solution at η_k is used as the initial condition for η_{k+1}, dramatically reducing the number of pseudo-transient steps required.

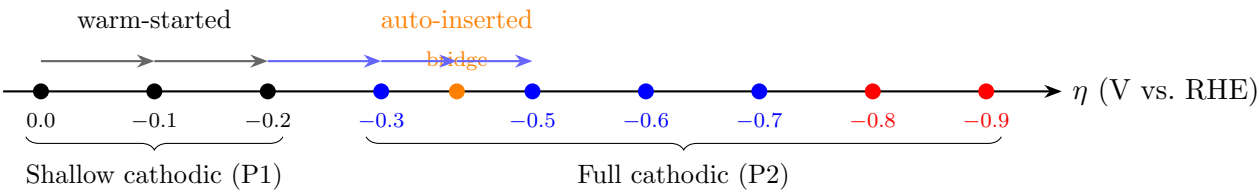

Figure 2: Voltage continuation strategy. Solutions propagate left to right along the η-axis. Bridge points (orange) are auto-inserted when the gap between consecutive targets exceeds a threshold. The shallow cathodic range (P1, 10 points) provides warm-starting for the full cathodic range (P2, 15 points).

Bridge point auto-insertion. When the gap between consecutive voltage points exceeds a threshold, intermediate *bridge points* are automatically inserted to maintain continuity of the solution manifold. These bridge points are solved but not included in the output I–V curve.

Mixed-sign sweep ordering. Positive and negative overpotentials are handled by sweeping from equilibrium ($\eta = 0$) outward in both directions, reusing the equilibrium solution as the branch point.

Predictor warm-starts. Between voltage points in the continuation sweep, we use 3-point quadratic Lagrange interpolation to predict the solution at the next voltage:

$$U_{\text{pred}}(\eta_{k+1}) \approx \sum_{j=0}^{2} U(\eta_{k-j}) \prod_{\substack{m=0 \\ m \neq j}}^{2} \frac{\eta_{k+1} - \eta_{k-m}}{\eta_{k-j} - \eta_{k-m}}. \tag{15}$$

The predictor is validated: if the predicted concentration is negative or the potential exceeds physical bounds, the predictor is rejected and the previous converged solution is used instead.

3.4 Charge Continuation

For charged species, the Debye parameter $\hat{\varepsilon} = (\lambda_D / L_{\text{ref}})^2 \approx 9 \times 10^{-8}$ introduces extreme stiffness in the Poisson equation. We employ a two-phase charge continuation strategy:

1. **Phase 1 (Neutral sweep):** Set all charges $z_i = 0$, effectively decoupling the Poisson equation. The system reduces to pure diffusion-reaction, which is much easier to solve. Sweep the full voltage range to obtain approximate concentration profiles.

2. **Phase 2 (Charge ramping):** At the target overpotential, gradually restore the physical charges by ramping a factor $\zeta \in [0, 1]$:

$$z_i^{(\zeta)} = \zeta \cdot z_i^{\text{phys}}, \qquad \zeta = 0, 0.1, 0.2, \ldots, 1.0. \tag{16}$$

   The Poisson coupling strengthens progressively, with each step warm-started from the previous converged solution.

3.5 Convergence Safeguards

Several safeguards ensure robust convergence:

- **Exponent clipping:** The Butler–Volmer exponent $\alpha_j \hat{\eta}$ is clipped to $[-50, +50]$ to prevent floating-point overflow ($e^{50} \approx 5.2 \times 10^{21}$).
- **Concentration floor:** Surface concentrations in the BV expression are regularized via $\hat{c}_{\text{surf}} = \max(\hat{c}_i, \epsilon_c)$ with $\epsilon_c = 10^{-8}$, preventing $\log(0)$ in steric terms.
- **MUMPS auto-scaling:** The MUMPS direct solver is configured with `icntl(8)=77` for automatic matrix scaling, critical for the ill-conditioned systems arising from charged electrolytes.
- **Conservative line search:** The Newton line search maximum step is limited to $\lambda_{\max} = 0.5$, preventing overshoot in the highly nonlinear regime.
- **Jacobian lag:** The Jacobian is recomputed at every Newton step (no lagging) to maintain quadratic convergence near the solution.

The SNES configuration is summarized in Table 5.

Table 5: PETSc/SNES solver configuration.

Parameter	Value
SNES type	`newtonls`
Max iterations	200 (neutral) / 300 (charged)
Absolute tolerance	10^{-7}
Relative tolerance	10^{-10}
Step tolerance	10^{-12}
Line search	`l2`, $\lambda_{\max} = 0.5$
Divergence tolerance	10^{12}
Linear solver	MUMPS direct (`preonly` + `lu`)
MUMPS scaling	`icntl(8)` = 77 (auto)
MUMPS memory	`icntl(14)` = 80

4 The Inverse Problem

4.1 Control Variables

The parameter vector to be recovered is:

$$\boldsymbol{\theta} = \left(\log_{10} k_0^{(1)},\ \log_{10} k_0^{(2)},\ \alpha_1,\ \alpha_2\right) \in \mathbb{R}^4. \tag{17}$$

Rate constants are parameterized in log-space because they span orders of magnitude ($k_0^{(2)}/k_0^{(1)} \approx 1/24$) and log-space provides natural scaling for the optimizer.

4.2 Objective Function

Given observed I–V curves $I_{\text{CD}}^{\text{obs}}(\eta_m)$ and $I_{\text{PC}}^{\text{obs}}(\eta_m)$ at voltage points $\{\eta_m\}_{m=1}^{M}$, the objective function is:

$$\mathcal{J}(\boldsymbol{\theta}) = \frac{1}{2}\sum_{m=1}^{M}\left(I_{\text{CD}}(\eta_m;\boldsymbol{\theta}) - I_{\text{CD}}^{\text{obs}}(\eta_m)\right)^2 + \frac{w}{2}\sum_{m=1}^{M}\left(I_{\text{PC}}(\eta_m;\boldsymbol{\theta}) - I_{\text{PC}}^{\text{obs}}(\eta_m)\right)^2, \tag{18}$$

where $w > 0$ is a weight controlling the relative importance of the peroxide current observable. The weight w is varied across pipeline phases to emphasize different observables (Section 6.4).

4.3 Gradient Computation

PDE phases. For the PDE-constrained optimization phases, gradients of $\mathcal{J}$ with respect to $\boldsymbol{\theta}$ are computed via the adjoint method. The forward PDE solve produces the state $U(\boldsymbol{\theta})$; two adjoint passes (one per observable) yield the gradient components. The adjoint equations are assembled and solved using dolfin-adjoint (pyadjoint).

The gradient in log-space follows from the chain rule:

$$\frac{\partial \mathcal{J}}{\partial \log_{10} k_0^{(j)}} = k_0^{(j)} \ln(10) \cdot \frac{\partial \mathcal{J}}{\partial k_0^{(j)}}. \tag{19}$$

Surrogate phases. For surrogate-based optimization, gradients are approximated by central finite differences with step size $h = 10^{-5}$:

$$\frac{\partial \mathcal{J}}{\partial \theta_i} \approx \frac{\mathcal{J}(\boldsymbol{\theta} + h\,\boldsymbol{e}_i) - \mathcal{J}(\boldsymbol{\theta} - h\,\boldsymbol{e}_i)}{2h}. \tag{20}$$

Each gradient evaluation requires $2 \times 4 = 8$ surrogate predictions, each costing $\sim$1 ms.

4.4 L-BFGS-B Optimizer

All optimization phases use the L-BFGS-B algorithm (limited-memory Broyden–Fletcher–Goldfarb–Shanno with box constraints), well-suited to this problem because:

- The objective is smooth (continuous PDE solution or smooth NN surrogate).
- Box constraints enforce physical bounds on $\alpha \in [0.1, 0.9]$ and $\log_{10} k_0 \in [-7, 0]$.
- The parameter dimension ($d = 4$) is low, so the L-BFGS Hessian approximation is highly effective.

Table 6: L-BFGS-B optimizer settings per pipeline phase.

Phase	`maxiter`	`ftol`	`gtol`
S1 (alpha init)	30	10^{-14}	10^{-8}
S2 (joint)	60	10^{-14}	10^{-8}
S3 (cascade, per pass)	60 / 60 / 30	10^{-14}	10^{-8}
S4 (multistart polish)	60	10^{-14}	10^{-8}
P1 (PDE shallow)	20	10^{-12}	10^{-6}
P2 (PDE full)	20	10^{-12}	10^{-6}

4.5 The $k_{0,2}$ Identifiability Challenge

The second rate constant $k_0^{(2)}$ is the most difficult parameter to recover for several reasons:

1. **Reaction rate ratio:** At moderate overpotentials, $\mathcal{R}_2 \ll \mathcal{R}_1$ because $k_0^{(2)}$ is 24× smaller. The contribution of R_2 to total current is marginal.
2. **Cancellation sensitivity:** The peroxide current $I_{\mathrm{PC}} \propto \mathcal{R}_1 - \mathcal{R}_2$ is a difference of two nearly equal quantities at large $|\eta|$, making it sensitive to numerical errors and noise amplification.
3. **Anti-correlation:** The optimizer can trade $k_0^{(1)}$ against $k_0^{(2)}$—increasing one while decreasing the other—and achieve a similar total current. This manifests as an elongated valley in the objective landscape.

Table 7 documents this trade-off across pipeline versions.

Table 7: The $k_0^{(1)}$–$k_0^{(2)}$ trade-off across versions. Prior to v13, no version achieved $< 10\%$ on both rate constants simultaneously.

Version	$k_0^{(1)}$ error	$k_0^{(2)}$ error	Sum	Max
v7 (PDE only)	10.9%	2.6%	13.5%	10.9%
v11 P4 (RBF→PDE)	9.4%	0.8%	10.2%	9.4%
v12 P3 (NN→PDE)	6.8%	11.8%	18.6%	11.8%
v12 P4 (NN→PDE full)	16.6%	6.0%	22.6%	16.6%
v13 P2	**4.78%**	**4.23%**	**9.01%**	**5.26%**

5 Surrogate Models

5.1 Training Data Generation

Training samples are generated via Latin Hypercube Sampling (LHS) over the four-dimensional parameter space:

- $\log_{10}(k_0^{(1)}) \in [-6, 0]$ (log-uniform),
- $\log_{10}(k_0^{(2)}) \in [-7, -1]$ (log-uniform),
- $\alpha_1 \in [0.1, 0.9]$ (uniform),
- $\alpha_2 \in [0.1, 0.9]$ (uniform).

Each sample requires a full PDE voltage sweep producing a 22-point I–V curve pair $(I_{\mathrm{CD}}, I_{\mathrm{PC}})$. Approximately 2000 samples are used for the NN ensemble and 445 for the RBF baseline.

5.2 Neural Network Architecture

We use a residual MLP (ResNet-MLP) with the architecture shown in Figure 3.

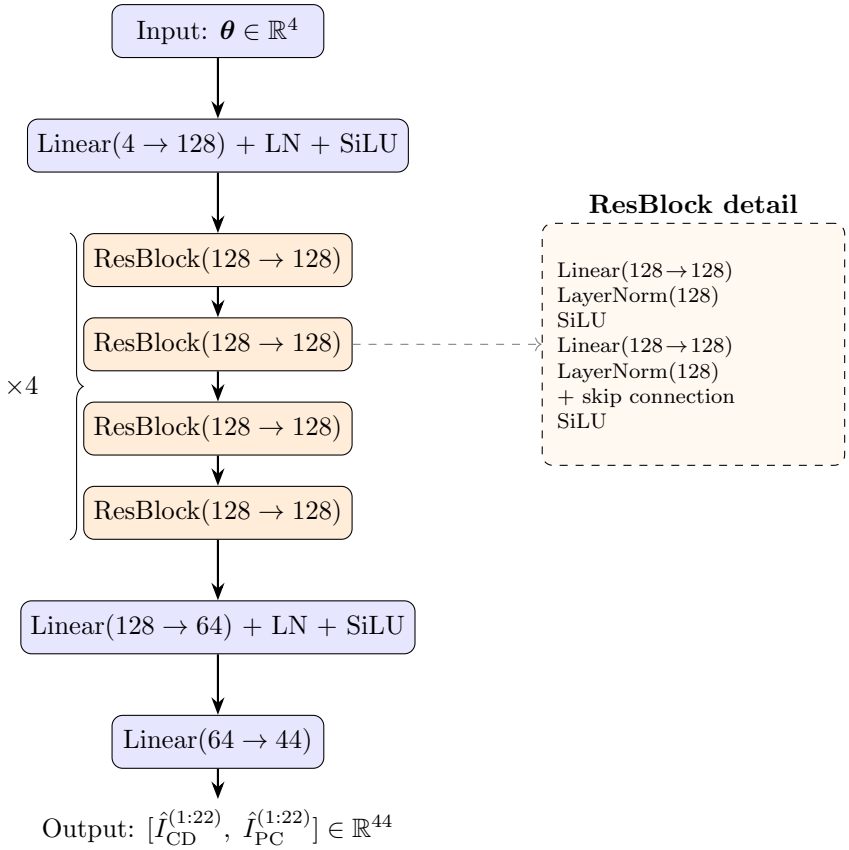

Figure 3: ResNet-MLP architecture. Four residual blocks with LayerNorm and SiLU activation map the 4-parameter input to 44 outputs (22 current density + 22 peroxide current values). Total: ∼145K parameters.

Architecture details. The input layer projects the 4-dimensional parameter vector to a 128-dimensional hidden representation via a linear layer, LayerNorm, and SiLU (Sigmoid Linear Unit, $\mathrm{SiLU}(x) = x\sigma(x)$) activation. Four residual blocks each apply two linear layers with LayerNorm and a skip connection. The output head compresses to 64 dimensions before a final linear layer produces the 44-dimensional output (22 voltage points × 2 observables).

5.3 Training Procedure

Table 8: Neural network training hyperparameters.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}
LR schedule	CosineAnnealingWarmRestarts ($T_0 = 500$, $T_{\mathrm{mult}} = 2$, $\eta_{\min} = 10^{-6}$)
Max epochs	5000
Early stopping patience	500 epochs
Validation fraction	15%
Loss	MSE (full-batch)
Input normalization	Z-score (k_0 in $\log_{10}$ space)
Output normalization	Z-score (concatenated CD+PC)

Training uses full-batch gradient descent with AdamW and a cosine annealing schedule with warm restarts. The warm restart periods double each cycle ($T_0 = 500$, $T_{\text{mult}} = 2$), allowing the optimizer to escape shallow local minima. Early stopping monitors validation loss with a patience of 500 epochs, restoring the best model.

Input normalization applies z-score standardization after transforming rate constants to $\log_{10}$ space. Output normalization z-scores the concatenated 44-dimensional curve vector. Standard deviations are clamped at 10^{-15} for numerical stability.

5.4 Ensemble Construction

An ensemble of $K = 5$ independently trained networks provides both improved predictions and uncertainty quantification:

- Each member uses a different random seed for weight initialization and a different train/-validation split (permutation-based).
- The ensemble prediction is the member-wise mean: $\bar{f}(\boldsymbol{\theta}) = \frac{1}{K}\sum_{k=1}^{K} f_k(\boldsymbol{\theta})$.
- Pointwise uncertainty is estimated by the ensemble standard deviation: $\sigma(\boldsymbol{\theta}) = \text{std}_{k=1}^{K} f_k(\boldsymbol{\theta})$.

Ensembling improves over single models by averaging out individual network biases and providing a model-free uncertainty estimate. The ensemble spread is used diagnostically in Section 7 to explain why $k_0^{(2)}$ is harder to recover.

5.5 RBF Baseline

As a baseline, we fit two thin-plate spline RBF interpolants (one per observable) using `scipy.interpolate.RBFI`

- Kernel: thin-plate spline ($r^2 \log r$),
- Polynomial degree: 1,
- Smoothing: 0 (exact interpolation),
- Input normalization: z-score (with k_0 in $\log_{10}$ space).

The RBF model is simpler (no training loop, no hyperparameters beyond the kernel) but scales poorly with sample count ($O(N^3)$ construction) and provides no built-in uncertainty estimate.

5.6 Model Evaluation

Both surrogates expose a unified prediction API:

- `predict(`$k_0^{(1)}, k_0^{(2)}, \alpha_1, \alpha_2$`)` returns current density and peroxide current arrays.
- `predict_batch` accepts arrays for vectorized evaluation (critical for the multi-start grid search).
- `predict_with_uncertainty` (ensemble only) returns mean and standard deviation.

6 Inference Pipeline Architecture

6.1 Overview

The v13 pipeline comprises seven phases organized in two tiers: five surrogate-based phases (S1–S5) for rapid global exploration, followed by two PDE-constrained phases (P1–P2) for high-accuracy refinement. Figure 4 shows the complete architecture.

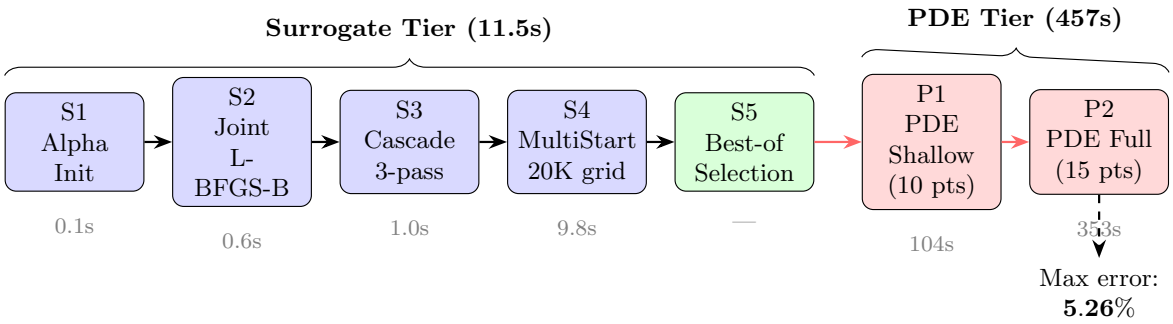

Figure 4: The seven-phase v13 inference pipeline. Surrogate phases (blue) perform rapid global exploration; the selection phase (green) chooses the best surrogate result; PDE phases (red) perform high-accuracy refinement. Timing annotations show wall-clock cost per phase.

6.2 Phase S1: Alpha Initialization

Purpose. Reduce early oscillation by initializing the transfer coefficients before joint optimization.

Method. Fix $k_0^{(1)}$ and $k_0^{(2)}$ at cold-guess values (e.g., from literature or order-of-magnitude estimates). Optimize α_1 and α_2 only using L-BFGS-B on the surrogate objective with 30 iterations.

Result. Produces a 2-parameter warm start for the transfer coefficients. Time: $\sim$0.1 s.

6.3 Phase S2: Joint Surrogate Optimization

All four parameters are optimized jointly via L-BFGS-B on a shallow voltage subset of the I–V curve. The surrogate provides fast function and gradient evaluations ($\sim$1 ms per call), enabling 60 iterations in 0.6 s.

This phase establishes an initial estimate in the correct basin for all four parameters, achieving 6.9% error on $k_0^{(1)}$ and 3.5% on both α values, but leaving $k_0^{(2)}$ at 11.9%.

6.4 Phase S3: Cascade Per-Observable Inference

The cascade strategy decouples reaction parameters through three sequential passes with different objective weightings:

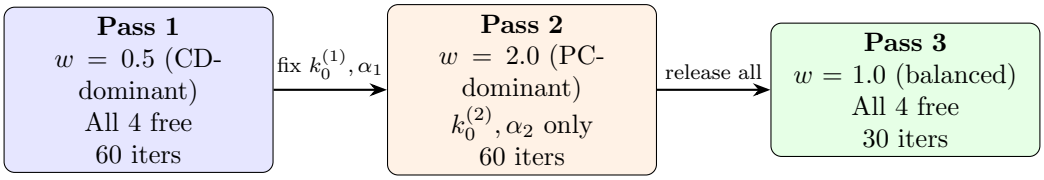

Figure 5: Cascade three-pass strategy. Pass 1 (CD-dominant) primarily recovers $k_0^{(1)}$ and α_1. Pass 2 (PC-dominant) focuses on $k_0^{(2)}$ and α_2 with reaction 1 parameters fixed. Pass 3 performs a balanced joint polish.

Rationale. When $w < 1$, the objective is dominated by total current density (sensitive to $k_0^{(1)}$). When $w > 1$, peroxide current (sensitive to $k_0^{(2)}$) is emphasized. By solving sequentially with increasing w, we first pin down reaction 1 parameters, then refine reaction 2 parameters against the peroxide signal, and finally polish all four jointly.

6.5 Phase S4: MultiStart Grid Search

Purpose. Verify global optimality of the surrogate landscape and guard against local minima.

Method.

1. Generate 20,000 LHS samples in the 4D parameter space (rate constants in log-space, transfer coefficients in linear space).
2. Evaluate all samples via vectorized batch prediction.
3. Select the top 20 candidates by objective value.
4. Polish each candidate with L-BFGS-B (60 iterations).
5. Return the best polished result.

Finding. All three strategies (S2, S3, S4) converge to the same optimum, confirming that the NN ensemble surrogate landscape has a *single global basin*. The multi-start adds a global optimality certificate but no accuracy improvement over S2.

6.6 Phase S5: Best-of Selection

Compare the surrogate objective values from S2, S3, and S4; select the winner. In our experiments, S4 (multi-start) achieves the marginally lowest loss (5.866×10^{-5}), though all three agree to 5 significant figures.

The ensemble uncertainty at the selected point is computed: mean CD standard deviation $\sigma_{\text{CD}} = 1.96 \times 10^{-4}$ and mean PC standard deviation $\sigma_{\text{PC}} = 9.78 \times 10^{-4}$. The 5× higher PC uncertainty reflects the ensemble's disagreement on the cancellation-sensitive peroxide observable and foreshadows the difficulty of recovering $k_0^{(2)}$.

6.7 Phases P1–P2: PDE Refinement

The surrogate optimum (S5 output) is used to warm-start two PDE-constrained optimization phases.

Phase P1: Shallow cathodic (10 voltage points). L-BFGS-B optimization with adjoint gradients on a voltage range covering moderate cathodic overpotentials. The reduced voltage range limits cost (104 s) while providing PDE-quality gradients that correct surrogate biases.

P1 serves primarily as a *basin selector*: it ensures the optimizer is in the correct region of parameter space before the expensive P2 phase. Ablation studies (Section 7.4) show that skipping P1 leads to 3.2× worse maximum error.

Phase P2: Full cathodic (15 voltage points). The final phase extends the voltage range to include deep cathodic overpotentials where the peroxide current provides maximal information about $k_0^{(2)}$. Starting from the P1 result, L-BFGS-B converges in 15 iterations (of a 20-iteration budget), requiring 353 s.

Key mechanism. At deep cathodic potentials, both reactions are strongly driven, and the ratio $\mathcal{R}_1/\mathcal{R}_2$ becomes more sensitive to $k_0^{(2)}$. The PDE's exact treatment of this ratio—unavailable to the surrogate's approximate function—enables P2 to break the $k_0^{(1)}$–$k_0^{(2)}$ anti-correlation and reduce $k_0^{(2)}$ error from 12% (surrogate) to 4.2%.

Algorithm 1: v13 Inference Pipeline

Input: Observed I–V curves $I_{\mathrm{CD}}^{\mathrm{obs}}$, $I_{\mathrm{PC}}^{\mathrm{obs}}$; trained NN ensemble $\mathcal{E}$; PDE forward solver $\mathcal{F}$
Output: Estimated parameters $\hat{\boldsymbol{\theta}}$

```
/* Surrogate tier                                                    */
```
$\boldsymbol{\theta}_0 \leftarrow$ cold guess
$\alpha_1^*, \alpha_2^* \leftarrow$ S1-AlphaInit$(\boldsymbol{\theta}_0, \mathcal{E})$
$\boldsymbol{\theta}_{\mathrm{S2}} \leftarrow$ S2-JointLBFGSB$(\boldsymbol{\theta}_0[\alpha \leftarrow \alpha^*], \mathcal{E})$
$\boldsymbol{\theta}_{\mathrm{S3}} \leftarrow$ S3-Cascade$(\boldsymbol{\theta}_{\mathrm{S2}}, \mathcal{E})$
$\boldsymbol{\theta}_{\mathrm{S4}} \leftarrow$ S4-MultiStart$(\mathcal{E})$
$\boldsymbol{\theta}_{\mathrm{S5}} \leftarrow \arg\min_{\boldsymbol{\theta} \in \{\boldsymbol{\theta}_{\mathrm{S2}}, \boldsymbol{\theta}_{\mathrm{S3}}, \boldsymbol{\theta}_{\mathrm{S4}}\}} \mathcal{J}_{\mathcal{E}}(\boldsymbol{\theta})$
```
/* PDE tier                                                          */
```
$\boldsymbol{\theta}_{\mathrm{P1}} \leftarrow$ P1-PDE-Shallow$(\boldsymbol{\theta}_{\mathrm{S5}}, \mathcal{F}, \eta_{\mathrm{shallow}})$
$\hat{\boldsymbol{\theta}} \leftarrow$ P2-PDE-Full$(\boldsymbol{\theta}_{\mathrm{P1}}, \mathcal{F}, \eta_{\mathrm{full}})$
return $\hat{\boldsymbol{\theta}}$

7 Results

All results use synthetic observations generated from the true parameters (Table 2) with 2% additive Gaussian noise (seed 20260226). Figure 6 shows the forward model I–V curves at the true parameter values, illustrating the two observables that the inverse problem seeks to match.

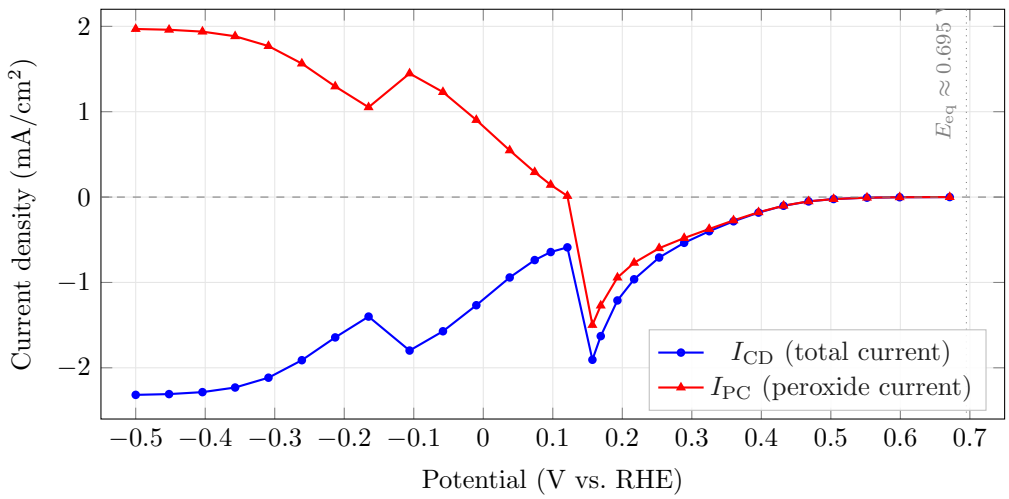

Figure 6: Forward model I–V curves at the true parameter values ($k_0^{(1)} = 2.4 \times 10^{-8}$ m/s, $k_0^{(2)} = 1.0 \times 10^{-9}$ m/s, $\alpha_1 = 0.627$, $\alpha_2 = 0.5$). Blue: total current density I_{CD} (analogous to disk current in RRDE). Red: peroxide current density I_{PC} (analogous to ring current). Near equilibrium (~ 0.7 V vs. RHE) both currents vanish; at cathodic potentials the total current approaches the diffusion-limited plateau while the peroxide current reflects the competition between R_1 (production) and R_2 (consumption). The non-monotonic feature in I_{PC} near 0.1–0.15 V arises from the transition where R_2 begins to dominate peroxide consumption.

7.1 Surrogate Phase Results

Table 9: Surrogate phase results (S1–S5). All percentage errors are relative to true parameter values.

Phase	Strategy	$k_0^{(1)}$	$k_0^{(2)}$	α_1	α_2	Max
S1	Alpha-only init	49.7%	78.5%	49.7%	78.5%	78.5%
S2	Joint L-BFGS-B	6.88%	11.85%	3.54%	3.48%	11.85%
S3	Cascade 3-pass	6.89%	12.24%	3.57%	3.43%	12.24%
S4	MultiStart 20K	6.89%	12.05%	3.55%	3.44%	12.05%
S5	**Best (S4)**	**6.89%**	**12.05%**	**3.55%**	**3.44%**	**12.05%**

Key finding. All three optimization strategies converge to essentially the same optimum (losses agree to 5 significant figures). This confirms a *single global basin* in the NN ensemble surrogate landscape. The cascade and multi-start add robustness (a global optimality certificate) but not accuracy. The surrogate plateau at $\sim$12% $k_0^{(2)}$ error represents the fundamental resolution limit of the NN approximation on this observable.

7.2 PDE Phase Results

Table 10: PDE refinement results (P1–P2).

Phase	Grid	$k_0^{(1)}$	$k_0^{(2)}$	α_1	α_2	Max
P1	Shallow (10 pts)	6.80%	12.01%	3.88%	3.17%	12.01%
P2	**Full (15 pts)**	**4.78%**	**4.23%**	**2.95%**	**5.26%**	**5.26%**

The P2 breakthrough. The transition from P1 to P2 produces the paper's central result: $k_0^{(2)}$ error drops from 12.0% to 4.23%—a 2.8× improvement. The extended voltage range provides deep-cathodic peroxide current data where Reaction 2 makes a non-negligible contribution, enabling the PDE solver to resolve the $k_0^{(1)}$–$k_0^{(2)}$ degeneracy.

Redimensionalized values.

Parameter	True	Estimated	Error
$k_0^{(1)}$	2.400×10^{-8} m/s	2.515×10^{-8} m/s	4.78%
$k_0^{(2)}$	1.000×10^{-9} m/s	9.577×10^{-10} m/s	4.23%
α_1	0.627	0.609	2.95%
α_2	0.500	0.474	5.26%

7.3 Historical Comparison

Table 11: Cross-version comparison of inference results at 2% noise.

Version	Architecture	$k_0^{(1)}$	$k_0^{(2)}$	α_1	α_2	Max
v7	PDE only	10.9%	2.6%	5.6%	8.8%	10.9%
v9	RBF surrogate	8.8%	7.6%	4.8%	6.4%	8.8%
v11	RBF→PDE	9.4%	0.8%	5.1%	7.8%	9.4%
v12	NN→PDE	6.8%	11.8%	3.9%	3.2%	11.8%
v13	**NN (multi)→PDE**	**4.78%**	**4.23%**	**2.95%**	**5.26%**	**5.26%**

v13 is the first version to achieve $< 10\%$ error on all four parameters simultaneously, and the first to break the $k_0^{(1)}$–$k_0^{(2)}$ trade-off (lowest sum at 9.01% and lowest max at 5.26%).

7.4 Ablation: Skip-P1

To assess whether Phase P1 is essential, we run an ablation that skips P1 and feeds the surrogate result (S5) directly to P2.

Table 12: Ablation: effect of skipping Phase P1.

Metric	v13 (with P1)	v13 (skip P1)
$k_0^{(1)}$ error	4.78%	16.99%
$k_0^{(2)}$ error	4.23%	6.15%
α_1 error	2.95%	8.24%
α_2 error	5.26%	11.78%
Max error	**5.26%**	16.99%
P2 time	353s	469s
Total time	541s	552s

Skipping P1 degrades all parameters and makes P2 paradoxically *slower* (the optimizer starts further from the PDE basin and requires more forward solves). P1 is not redundant; it is an essential basin selector.

7.5 Timing Breakdown

Table 13: Wall-clock timing breakdown for the v13 pipeline.

Stage	Time (s)	Fraction
Target generation (PDE at truth)	71.7	13.3%
S1: Alpha-only init	0.1	$< 0.1\%$
S2: Joint L-BFGS-B	0.6	0.1%
S3: Cascade (3 passes)	1.0	0.2%
S4: MultiStart (20K + polish)	9.8	1.8%
P1: PDE shallow cathodic	103.9	19.2%
P2: PDE full cathodic	353.3	65.3%
Total	**540.9**	**100%**

Surrogate phases S1–S4 collectively cost 11.5 s (2.1% of total), while PDE phases account for 84.5%. The surrogate's role is *warm-starting*, not accuracy: it provides a good initial guess that reduces PDE optimization cost by $\sim$30% compared to cold starts.

8 Computational Optimizations

8.1 Parallel PDE Evaluation

Each PDE phase requires evaluating the forward model at multiple voltage points. These evaluations are independent and can be parallelized using a process-based worker pool:

- Pool size: $\min(M, \texttt{cpu_count} - 1)$, where M is the number of voltage points.
- Workers are spawned via Python's `multiprocessing` module.
- Each worker imports Firedrake independently (no shared state).
- Results are gathered and assembled into I–V curves.

8.2 IC Cache and Warm-Starting

Warm-starting is the single most impactful optimization. Two cache levels are maintained:

1. **All-points cache:** Stores converged solutions for every (k_0, α, η) triple evaluated during optimization. Subsequent evaluations at the same η with nearby parameters start from the cached solution.
2. **Cross-evaluation cache:** Solutions from evaluation n warm-start evaluation $n + 1$, exploiting the fact that L-BFGS-B steps are small in parameter space.

Warm-started solves use `max_steps=20` pseudo-transient iterations, versus 50–100 for cold starts. This reduces per-evaluation cost by $\sim$60%.

8.3 Two-Tape Adjoint

The multi-observable objective (18) requires derivatives with respect to both I_{CD} and I_{PC}. Rather than two forward solves (one taped per observable), we use a single forward solve with two adjoint passes:

1. Tape the forward solve once.

2. Adjoint pass 1: differentiate I_{CD} w.r.t. $\boldsymbol{\theta} \to \nabla_{\boldsymbol{\theta}} J_{\mathrm{CD}}$.
3. Adjoint pass 2: differentiate I_{PC} w.r.t. $\boldsymbol{\theta} \to \nabla_{\boldsymbol{\theta}} J_{\mathrm{PC}}$.
4. Combine: $\nabla \mathcal{J} = \nabla J_{\mathrm{CD}} + w \, \nabla J_{\mathrm{PC}}$.

This halves the forward computation cost relative to a naive implementation.

8.4 Shared Pool Reuse

The worker pool is created once at the start of P1 and reused for P2. This saves $\sim 10\,\mathrm{s}$ of Firedrake import overhead per worker. Between phases, the solution cache is cleared to prevent stale warm-starts from misguiding the optimizer.

9 Discussion

9.1 Why $k_{0,2}$ Is the Bottleneck

The difficulty of recovering $k_0^{(2)}$ has three layers:

Physical. At moderate overpotentials ($|\eta| < 0.5\,\mathrm{V}$), Reaction 2's rate is $\sim 24\times$ smaller than Reaction 1's. The total current is insensitive to $k_0^{(2)}$, and the peroxide current $I_{\mathrm{PC}} = I_{\mathrm{R1}} - I_{\mathrm{R2}}$ is a difference of two nearly equal quantities—amplifying measurement noise.

Surrogate. The NN ensemble's uncertainty on peroxide current ($\sigma_{\mathrm{PC}} = 9.78 \times 10^{-4}$) is $5\times$ higher than on current density ($\sigma_{\mathrm{CD}} = 1.96 \times 10^{-4}$). The ensemble members disagree most where the signal-to-noise ratio is worst. This places a fundamental floor on surrogate-only recovery of $k_0^{(2)}$.

Voltage range. At deep cathodic potentials, both reactions saturate and I_{PC} becomes more sensitive to $k_0^{(2)}$. The transition from P1 (shallow) to P2 (full cathodic) provides exactly this information, enabling the 12% $\to$ 4.2% breakthrough.

9.2 The Continuation Analogy

The P1$\to$P2 transition can be understood as a continuation method in parameter space. P1 solves an *easier problem* (fewer voltage points, reduced landscape complexity) that nonetheless constrains the solution to a basin near the global optimum. P2 then refines within this basin using richer data.

This is analogous to classical homotopy methods in nonlinear optimization, where a sequence of problems $\mathcal{J}_\lambda$ ($\lambda \in [0, 1]$) interpolates between a tractable proxy and the true objective. Here, λ is implicitly controlled by the voltage range: more data $\Rightarrow$ harder but more informative problem.

The ablation (Table 12) quantifies the cost of skipping this continuation: without P1, P2 converges to a worse local minimum with $3.2\times$ higher maximum error.

9.3 Surrogate vs. PDE Roles

The pipeline embodies a *multi-fidelity paradigm*:

- **Surrogates for exploration:** Fast (1 ms per evaluation), approximate, used to identify the global basin and provide warm-starts. Three independent strategies confirm global optimality.

- **PDE for refinement:** Slow (20 s per evaluation), exact, used for final parameter recovery where surrogate accuracy is insufficient (particularly for $k_0^{(2)}$).

The surrogate phases contribute only 2% of total cost yet are essential: without them, PDE phases would require cold starts and potentially converge to inferior local minima.

9.4 Limitations

Several limitations should be noted:

1. **Single noise seed:** Results are reported for one noise realization (seed 20260226). Robustness across noise seeds has not been verified.
2. **Specific geometry:** The 2D rectangle with $L_{\text{ref}} = 100\,\mu\text{m}$ is specific to the RDE configuration. Extension to 3D or different geometries requires retraining surrogates.
3. **Noise level:** Only 2% Gaussian noise has been tested. At 0% noise, v11 achieved 1.07% maximum error, suggesting that noise is the dominant error source—a 9× degradation.
4. **Surrogate training cost:** The ∼2000 PDE solves for training data generation (∼10 h) are a one-time cost not included in the pipeline timing.

10 Conclusions

We presented a seven-phase multi-fidelity inference pipeline that recovers four Butler–Volmer kinetic parameters from noisy I–V curve measurements with $< 5.3\%$ maximum error. The key results are:

1. **Breaking the trade-off:** v13 is the first pipeline version to achieve $< 10\%$ error on all four parameters simultaneously, resolving the persistent $k_0^{(1)}$–$k_0^{(2)}$ anti-correlation that limited prior versions.
2. **Multi-fidelity architecture:** The combination of NN ensemble surrogates (for global exploration) with PDE-constrained optimization (for local refinement) is more effective than either approach alone. Surrogates provide basin identification at 2% of total cost; PDE phases provide the accuracy.
3. **Continuation is essential:** The shallow-voltage PDE phase (P1) serves as a basin selector for the full-cathodic phase (P2). Skipping P1 degrades maximum error by 3.2×.
4. **Voltage range matters:** Deep cathodic overpotentials provide the peroxide current signal needed to resolve $k_0^{(2)}$, explaining the 12% → 4.2% improvement from P1 to P2.

Noise robustness remains an open challenge. The 5.26% headline result was obtained with a single noise seed. Preliminary experiments across five independent 2%-noise realisations reveal substantial seed-to-seed variability: the median maximum error rises to $\approx 23\%$ for both the NN-ensemble and POD-RBF-log surrogates, with only one out of five seeds achieving $<$ 10% maximum error for either model. We additionally compared four surrogate architectures (NN ensemble, thin-plate-spline RBF, and two POD-RBF variants) in a surrogate-only setting (no PDE refinement) and found that the noise-free ranking is *inverted* under noise: the RBF interpolant, which is the most accurate at 0% noise (0.27% max error), drops to second place at 2% noise (22.7% mean), while the NN ensemble's implicit smoothing makes it the most noise-robust (19.7% mean). This bias–variance trade-off, combined with the observation that

one adversarial noise seed drives *all* models above 55% error, indicates that surrogate warm-starting alone is insufficient for robust recovery. Because the PDE refinement phases (P1–P2) are local optimisers, their success depends critically on the quality of the initial guess supplied by the surrogate. When noise corrupts the surrogate's parameter estimate—particularly $k_0^{(2)}$, which sits in a shallow, poorly resolved region of the loss landscape—the PDE optimiser inherits a starting point far from the true basin and cannot recover. This suggests that the primary bottleneck is not the PDE solver itself but the surrogate's ability to provide a reliable warm-start under noisy observations. Improved surrogate training—larger and more diverse training sets, noise-aware data augmentation, or architectures with better extrapolation properties—could therefore yield substantial gains in end-to-end robustness without modifying the PDE refinement stages.

Beyond surrogate improvements, future work includes regularisation strategies for the PDE objective (Tikhonov penalties, restricted voltage ranges), ensembling over multiple noise seeds, exploration of alternative surrogate architectures (e.g., physics-informed neural operators), and extension to three-dimensional electrode geometries.

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